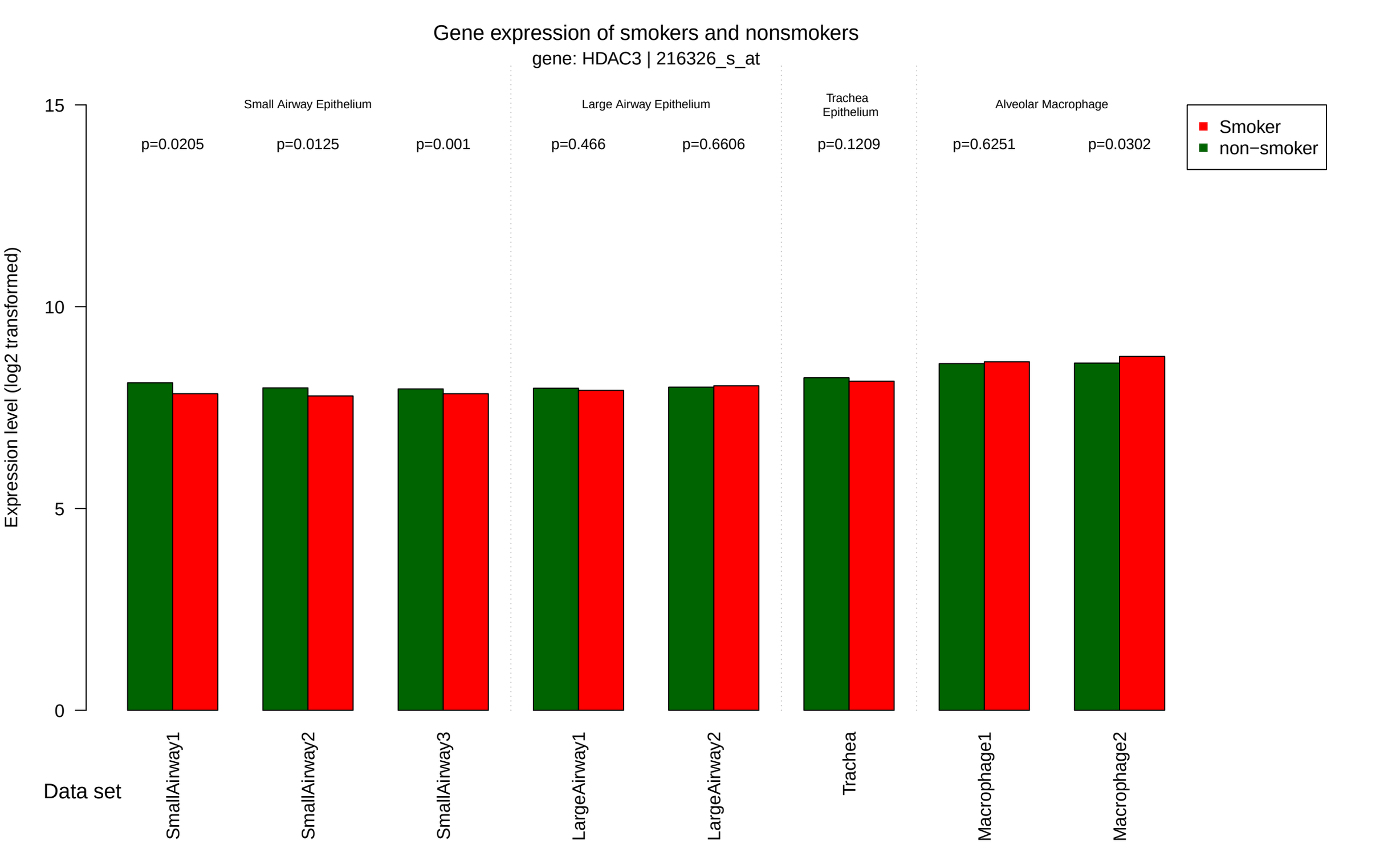
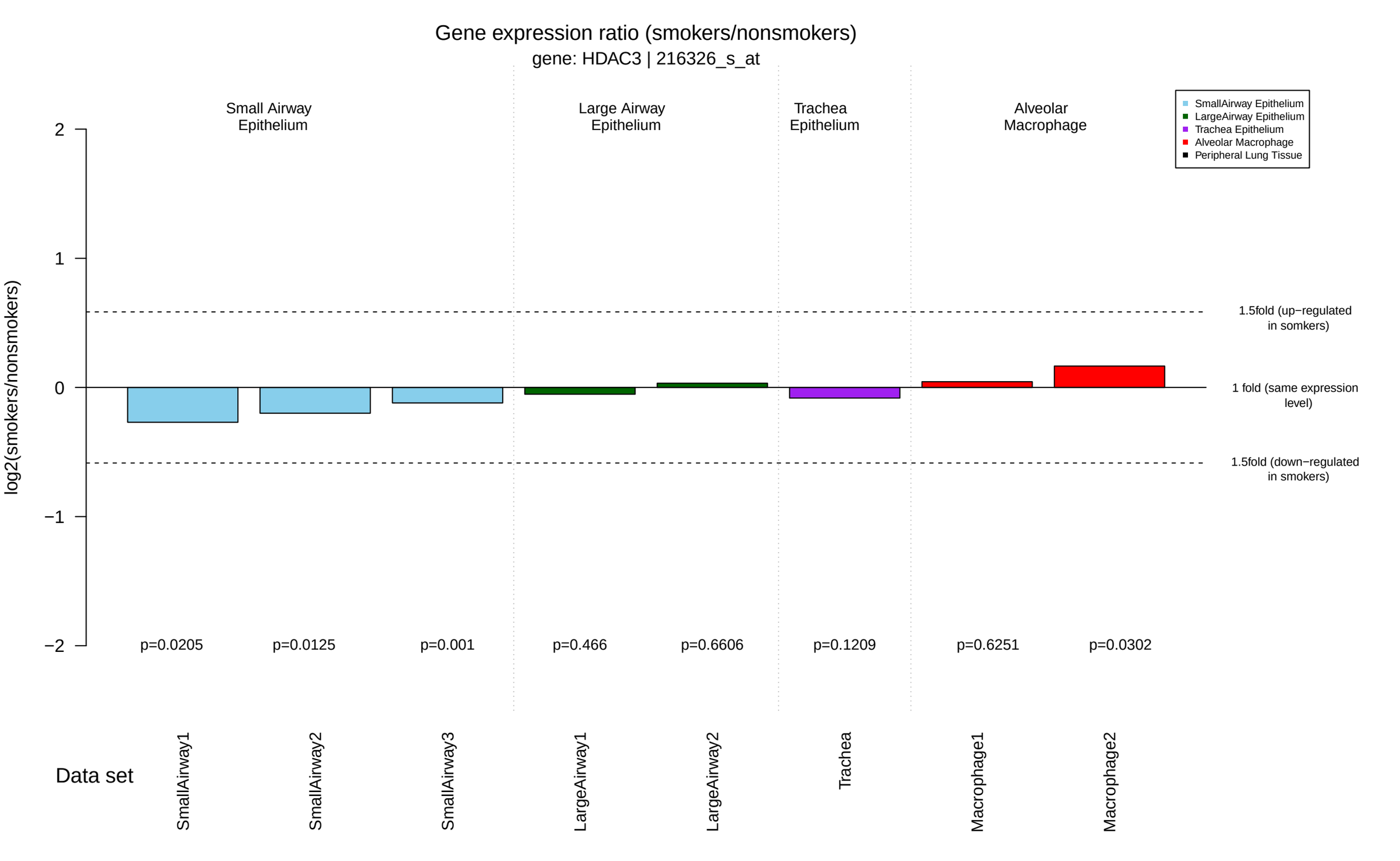
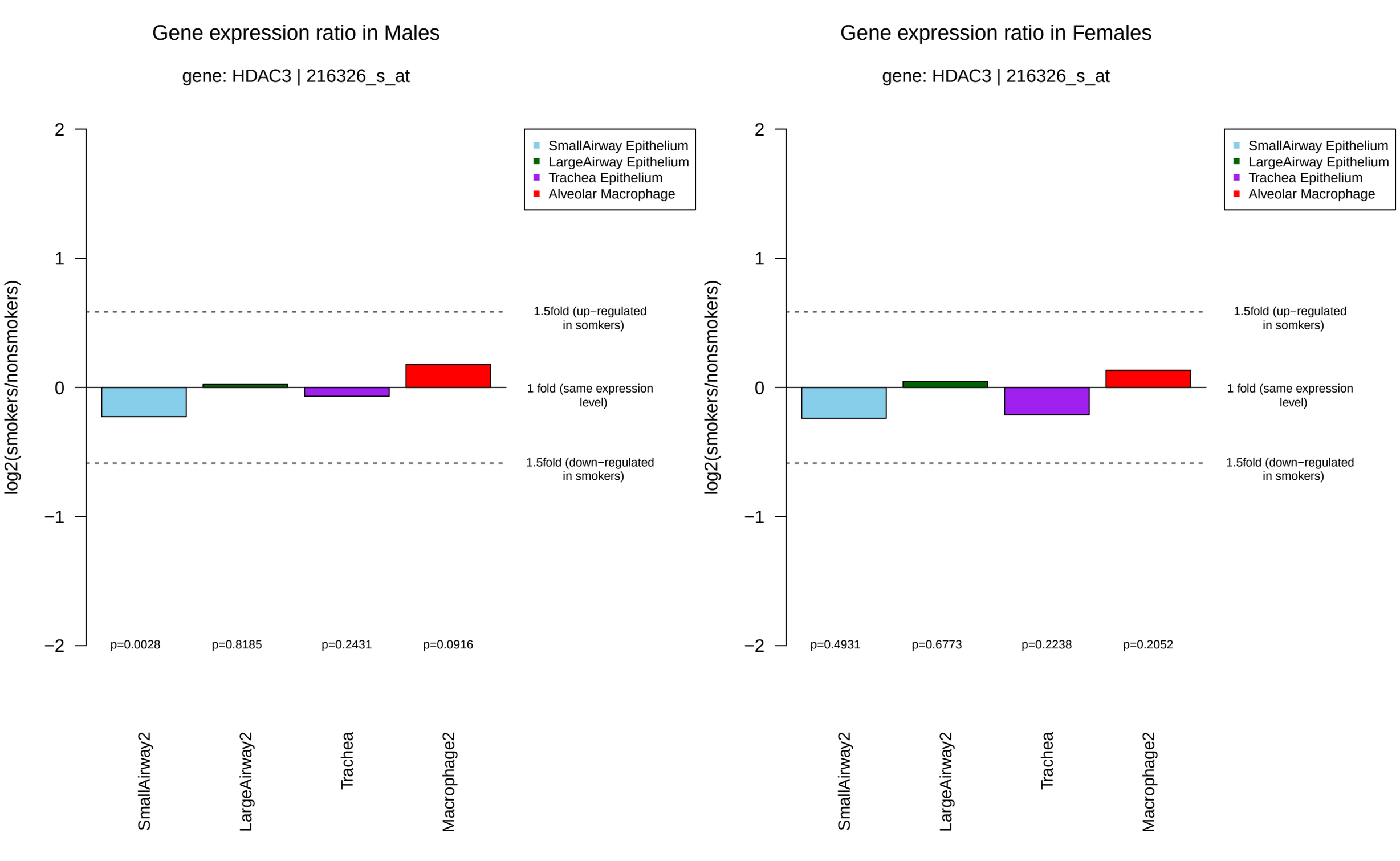
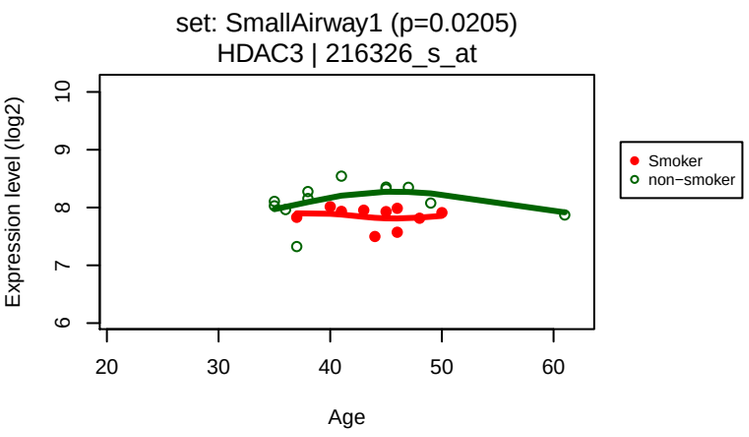
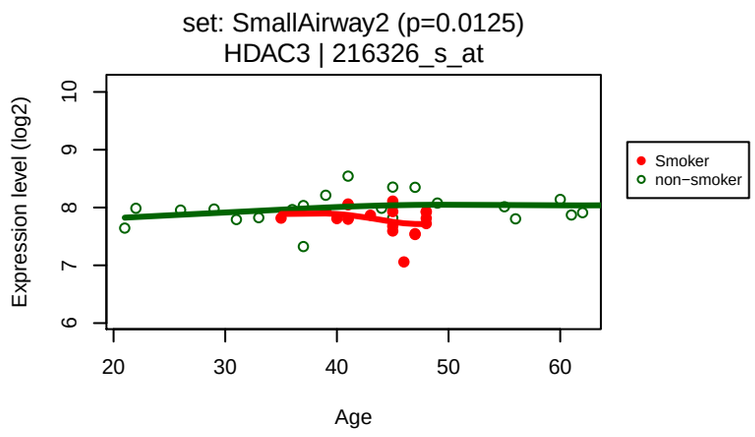
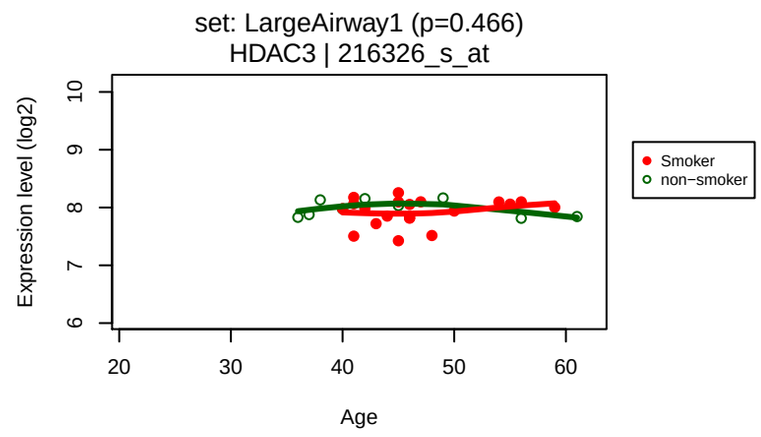

Gene expression of smokers and nonsmokers
gene: HDAC3 | 216326_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0205
p=0.0125
p=0.001
p=0.466
p=0.6606
p=0.1209
p=0.6251
p=0.0302
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

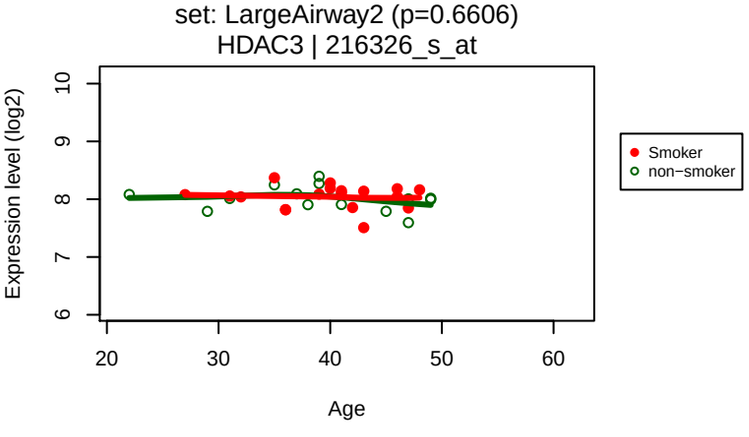

Gene expression ratio (smokers/nonsmokers)
gene: HDAC3 | 216326_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0205
p=0.0125
p=0.001
p=0.466
p=0.6606
p=0.1209
p=0.6251
p=0.0302
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

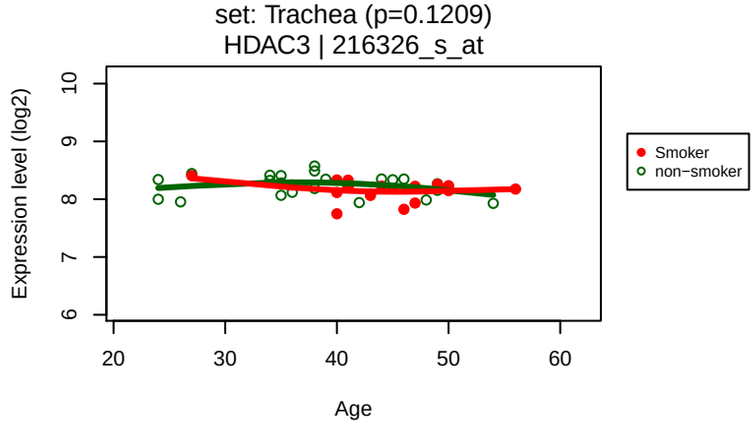

Gene expression ratio in Males
gene: HDAC3 | 216326_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0028
p=0.8185
p=0.2431
p=0.0916
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC3 | 216326_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4931
p=0.6773
p=0.2238
p=0.2052
SmallAirway2
LargeAirway2
Trachea
Macrophage2

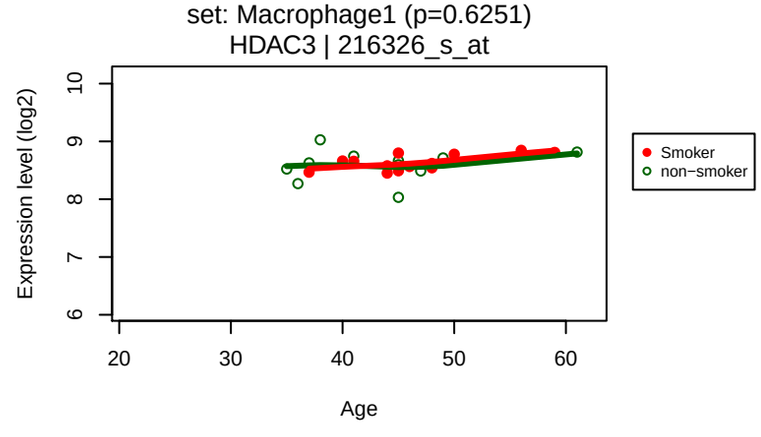

set: SmallAirway1 (p=0.0205)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker

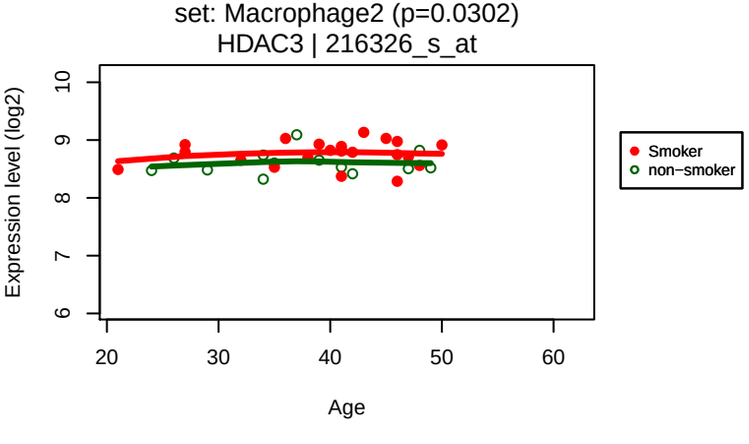

set: SmallAirway2 (p=0.0125)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker

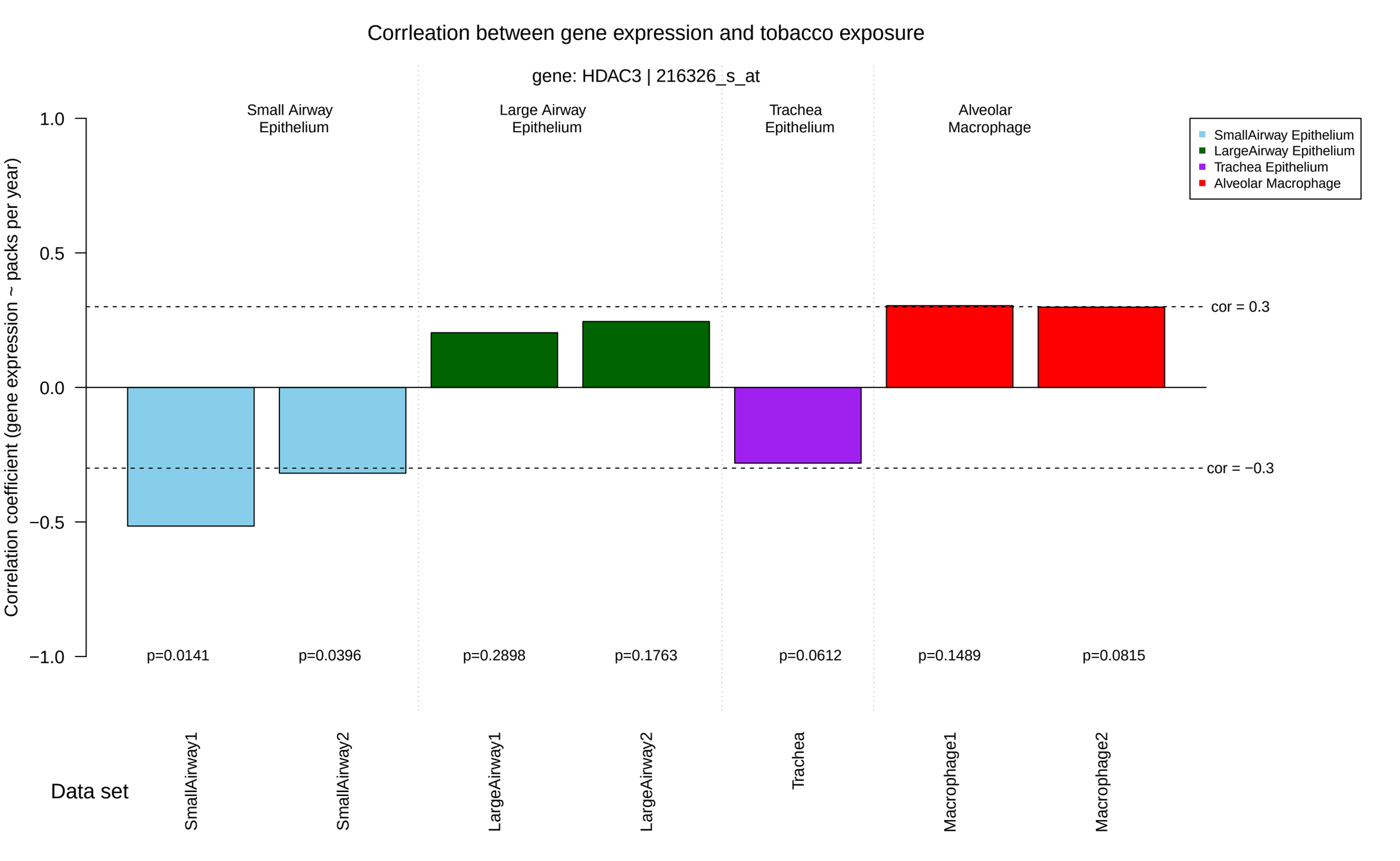

set: LargeAirway1 (p=0.466)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6606)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1209)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6251)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0302)
HDAC3 | 216326_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC3 | 216326_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0141
p=0.0396
p=0.2898
p=0.1763
p=0.0612
p=0.1489
p=0.0815
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2